Interiority in Optimal Transport with Congestion and Penalized Marginals

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October 3, 2026

Abstract

We give sufficient conditions for strictly positive optimal plans in the finite-dimensional transport model with quadratic congestion and penalized marginals of Gallardo, Loaiza, and Chávez. A uniform condition in the model primitives follows from the Karush–Kuhn–Tucker conditions. A refinement provides bounds for individual entries and can be checked in $O(NL)$ operations. Exact examples show that the refined condition is less restrictive and that neither condition is necessary. These results partially address the question of identifying explicit parameter regions where the interior linear-system representation is valid.

Keywords: Optimal transport; congestion; penalized marginals; interior solutions.
MSC: 90C20, 90C25, 91B68.

1 Introduction

The expanding use of artificial intelligence in mathematics motivates revisiting focused questions whose answers admit direct proofs. Examples include machine-learning-guided discoveries (Davies et al., 2021) and the recent announcement of an AI-generated proof of finite-time singularity formation for the three-dimensional Navier–Stokes equations with smooth external forcing (OpenAI, 2026). Motivated by these developments, we examine an open question in the transport model of Gallardo et al. (2025): which conditions on its primitives guarantee an interior optimal plan?

The model combines heterogeneous quadratic congestion costs with penalties for deviations from marginal targets. Its interior solutions are characterized by an unconstrained linear system, but positivity of that system's solution requires a separate argument. We provide a uniform sufficient condition and a refinement giving bounds for each entry. Both are elementary consequences of the KKT conditions and offer a partial answer to the question raised by Gallardo et al. (2025). Throughout, interiority means strict positivity in a finite-dimensional Euclidean orthant.

2 An interiority criterion

Let $N, L \geq 1$ be integers and $a_{ij}, c_{ij}, \epsilon_i, \delta_j, \mu_i, \nu_j > 0$. For $\alpha = 1/2$ in the model of Gallardo et al. (2025), consider

$$\min_{\pi \in \mathbb{R}_+^{N \times L}} F(\pi), \qquad \mathbb{R}_+ = [0, \infty), \tag{1}$$

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where, omitting fixed additive costs,

$$F(\pi) = \frac{1}{2}\sum_{i,j}(a_{ij}\pi_{ij}^2 + c_{ij}\pi_{ij}) + \frac{1}{2}\sum_i \epsilon_i \left(\sum_j \pi_{ij} - \mu_i\right)^2 + \frac{1}{2}\sum_j \delta_j \left(\sum_i \pi_{ij} - \nu_j\right)^2. \tag{2}$$

No equality between the total target masses is required. Set

$$b_{ij} = \epsilon_i\mu_i + \delta_j\nu_j - \frac{c_{ij}}{2}, \qquad h_{ij} = a_{ij} + \epsilon_i + \delta_j, \tag{3}$$

and write $b_{\min} = \min_{i,j} b_{ij}$, $b_{\max} = \max_{i,j} b_{ij}$, $a_{\min} = \min_{i,j} a_{ij}$, $\epsilon_{\max} = \max_i \epsilon_i$ and $\delta_{\max} = \max_j \delta_j$.

Theorem 2.1. *If*

$$b_{\min} > 0, \qquad a_{\min}b_{\min} > b_{\max}\big[(L-1)\epsilon_{\max} + (N-1)\delta_{\max}\big], \tag{4}$$

then (1) has a unique, strictly positive minimizer $\pi^\star$. More precisely, defining

$$M = \frac{b_{\max}}{a_{\min}}, \qquad \eta = b_{\min} - M\big[(L-1)\epsilon_{\max} + (N-1)\delta_{\max}\big] > 0,$$

we have $\eta/h_{ij} \le \pi^\star_{ij} \le M$ for every pair.

Proof. On the feasible set, $F(\pi) \ge (a_{\min}/2)\,\|\pi\|_F^2$. Hence continuity and coercivity give existence, and the positive quadratic coefficients a_{ij} give strict convexity and uniqueness.

For the constraints $-\pi_{ij} \le 0$, the Lagrangian is $\mathscr{L} = F - \sum_{i,j}\lambda_{ij}\pi_{ij}$, with $\lambda_{ij} \ge 0$. The all-ones matrix is strictly feasible, so the KKT conditions are necessary and sufficient (Boyd and Vandenberghe, 2004, Section 5.5.3):

$$\pi^\star_{ij} \ge 0, \qquad g_{ij} := \partial_{ij}F(\pi^\star) \ge 0, \qquad \pi^\star_{ij}g_{ij} = 0. \tag{5}$$

Differentiating and separating the (i,j) entry from its row and column gives

$$g_{ij} = h_{ij}\pi^\star_{ij} + \epsilon_i\sum_{\ell\neq j}\pi^\star_{i\ell} + \delta_j\sum_{k\neq i}\pi^\star_{kj} - b_{ij}. \tag{6}$$

If $\pi^\star_{ij} > 0$, then $g_{ij} = 0$ and the other entries are nonnegative, so $h_{ij}\pi^\star_{ij} \le b_{ij} \le b_{\max}$. If $\pi^\star_{ij} = 0$, the bound $\pi^\star_{ij} \le M$ is immediate. Since $h_{ij} \ge a_{\min}$, every entry therefore satisfies $0 \le \pi^\star_{ij} \le M$, without assuming interiority.

If some $\pi^\star_{ij} = 0$, equation (6) would imply

$$g_{ij} \le M\big[(L-1)\epsilon_i + (N-1)\delta_j\big] - b_{ij} \le -\eta < 0,$$

contradicting (5). Thus every entry is positive. Substituting $g_{ij} = 0$ in (6) and using the same bounds gives $h_{ij}\pi^\star_{ij} \ge \eta$. Empty sums are zero, so the proof includes $N = 1$ or $L = 1$. □

3 A refinement

The quantity b_{ij} is the negative marginal cost at the zero plan. Other entries in its row and column increase that marginal cost. Bounding these interactions separately improves the uniform criterion. Write $t_+ = \max\{t, 0\}$.

Proposition 3.1. *Without imposing (4) or a sign on b_{ij}, define*

$$u_{ij} = \frac{(b_{ij})_+}{h_{ij}}, \qquad m_{ij} = b_{ij} - \epsilon_i\sum_{\ell\neq j}u_{i\ell} - \delta_j\sum_{k\neq i}u_{kj}. \tag{7}$$

Then $(m_{ij})_+/h_{ij} \le \pi^\star_{ij} \le u_{ij}$. Thus $m_{ij} > 0$ for every pair guarantees interiority. Condition (4) implies this refined criterion.

Proof. Existence, uniqueness and (5) hold independently of (4). Combining (5) and (6),

$$\pi^\star_{ij} = \frac{1}{h_{ij}} \left(b_{ij} - \epsilon_i \sum_{\ell \neq j} \pi^\star_{i\ell} - \delta_j \sum_{k \neq i} \pi^\star_{kj} \right)_+ .$$

The nonnegative subtracted terms give $\pi^\star_{ij} \leq u_{ij}$. Substituting these upper bounds and using monotonicity of the positive part gives the lower bound. Under (4), $u_{ij} \leq M$, so $m_{ij} \geq \eta > 0$. □

Computing $U_i = \sum_j u_{ij}$ and $V_j = \sum_i u_{ij}$ gives $m_{ij} = b_{ij} - \epsilon_i(U_i - u_{ij}) - \delta_j(V_j - u_{ij})$ in $O(NL)$ operations. Even when the full criterion fails, $m_{ij} > 0$ certifies a positive entry, while $b_{ij} \leq 0$ forces that entry to vanish.

Remark 3.2. Let $H = \nabla^2 F$. Its positive diagonal cost term and positive semidefinite penalty terms make H positive definite. Since $\nabla F(\pi) = H\pi - b$, the optimum is interior if and only if $H^{-1}b > 0$ entrywise. Our criteria ensure this property without solving the system.

Remark 3.3. For a general weight $0 < \alpha < 1$, divide the original objective, with weight α on direct costs and $1 - \alpha$ on penalties, by $2(1 - \alpha)$. All results then apply with a_{ij}, c_{ij} replaced by $\kappa a_{ij}, \kappa c_{ij}$, where $\kappa = \alpha/(1 - \alpha)$.

4 Examples

Take $N = L = n$, $a_{ij} = A > 0$, $c_{ij} = 2$, and $\epsilon_i = \delta_j = \mu_i = \nu_j = 1$. Then $b_{ij} = 1$, $h_{ij} = A + 2$, and the constant plan $\pi^\star_{ij} = 1/(A + 2n)$ has zero gradient, hence is the unique optimum. The uniform and refined margins are $A - 2(n - 1)$ and $m_{ij} = 1 - 2(n - 1)/(A + 2)$, respectively:

n	A	Uniform margin	Refined margin	$\pi^\star_{ij}$
2	4	2	2/3	1/8
2	1	−1	1/3	1/5
3	1	−3	−1/3	1/7

The rows show, respectively, that the uniform condition is nonvacuous, the refinement is strictly less restrictive, and neither condition is necessary.

Positivity of b alone is insufficient. Set $N = 1$, $L = 2$, $a_{11} = a_{12} = \epsilon_1 = \delta_1 = \delta_2 = 1$, $\mu_1 = \nu_1 = \nu_2 = 3$, and $(c_{11}, c_{12}) = (10, 4)$. Then

$$H = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \quad \pi^\star = \begin{pmatrix} 0 \\ 4/3 \end{pmatrix}, \quad H\pi^\star - b = \begin{pmatrix} 1/3 \\ 0 \end{pmatrix}.$$

The KKT conditions certify this boundary optimum despite $b > 0$. Thus the sufficient criteria identify explicit interior regions without providing a complete classification.

AI statement

The authors used OpenAI's GPT-6 Astra with the ultra reasoning setting to assist in verifying algebraic calculations and proof logic, and in preparing the exposition. This is AI-assisted checking, not formal verification by a proof assistant. The authors remain responsible for the mathematical content.

Data availability. No empirical dataset is used. All example parameters and exact solutions are given above.

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